

QM-2
Oct-2

Today: "Periodic Boundary Conditions"

Photoelectric effect (5-8.3)

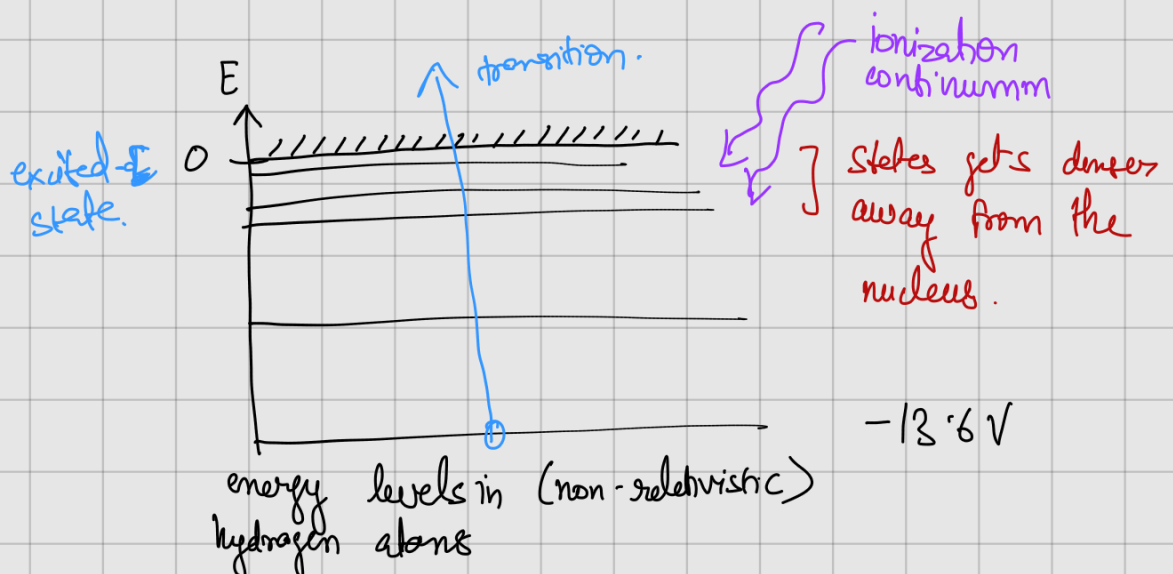
WARM UP: Dipole selection rules.

what polarization (q) of photon needed for
 $\psi_{100} \rightarrow \psi_{211}$

$\Rightarrow q = 1$
rule used: $m_f = m_i + q$

Photoelectric effect

- light $\rightarrow$ matter
 $\hookrightarrow$ emits electrons
- ion: Photo ionisation



transition rate :

$$W_{i \rightarrow \text{cont.}} = \frac{2\pi}{\hbar} |V_{fi}|^2 g_f(E_f)$$

Fermi's golden rule

electron is absorbing energy provided by photon.

→ for absorption, Fermi's golden rule is calculated at

$$E_f = E_i + \hbar\omega$$

$$W_{i \rightarrow \text{cont.}} = \frac{2\pi}{\hbar} |V_{fi}|^2 g_f(E_f = E_i + \hbar\omega)$$

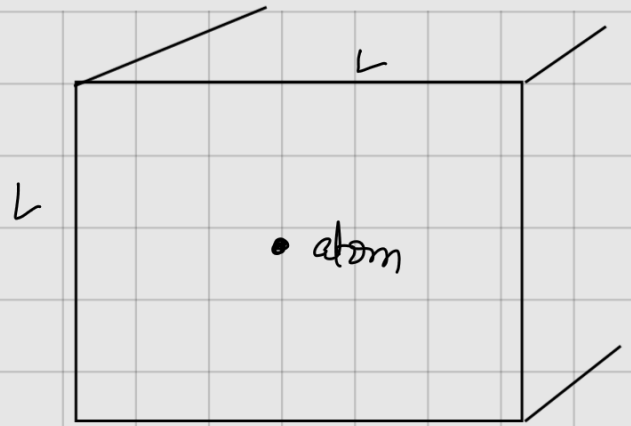
$$V = V^+ e^{-i\omega t} + V^- e^{i\omega t}$$

Density of states:

$$g(E) = \frac{dN}{dE} = \frac{\text{num. states}}{\text{energy}}$$

* we need to discretize the continuum:

Finite volume



atom in a box
of side L .

Fourier Series

$$\psi(\vec{r}) = \sum_{n_x, n_y, n_z} c_{\vec{n}} e^{i \frac{2\pi \vec{n} \cdot \vec{r}}{L}}$$

$$\vec{n} = (n_x, n_y, n_z)$$

$$n_i \in \mathbb{Z}$$

→ momentum eigenstates

$$\langle \vec{r} | \vec{k} \rangle \propto e^{i \vec{k} \cdot \vec{r}}$$

$$\Rightarrow \vec{k} = \frac{2\pi}{L} \vec{n}$$

$$\psi_{\vec{k}}(\vec{r}) = A e^{i \vec{k} \cdot \vec{r}}$$

Normalize:

$$1 = \int_{L^3} |\psi(\vec{r})|^2 d^3 \vec{r} = A \int_{L^3} 1 d^3 \vec{r}$$

$$1 = A L^3$$

$$\Rightarrow A = \frac{1}{L^{3/2}}$$

$$\psi_{\vec{k}}(\vec{r}) = \frac{1}{L^{3/2}} e^{i\vec{k} \cdot \vec{r}}, \quad \vec{k} = \frac{2\pi\vec{n}}{L}$$

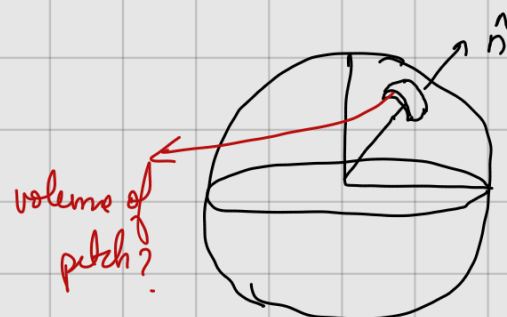
kinda like
particle in
a box, but
not really b/c
the box we
have the
atom in a
box which can
be as big
as we want
to be.

Density of states:

$$g = \frac{dN}{dE}$$

given $E \rightarrow$ shell in $\vec{n}$ space
given $E, dE \rightarrow$ shell, thickness dn

$\vec{n}, d\Omega \rightarrow$ patch on the shell



volume $\approx$ number of states ($\vec{n}$)

volume = area $\times$ thickness

$$= \underbrace{(n^2 d\Omega)}_{\text{area of the patch}} \underbrace{dn}_{\text{thickness}}$$

$$E = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2}{2m} \left(\frac{2\pi}{L} \right)^2 n^2 \equiv \alpha n^2$$

$$dE = 2\alpha n dn \rightarrow n dn = \frac{dE}{2\alpha}$$

$$\Rightarrow dE = 2\alpha n dn \rightarrow n dn = \frac{dE}{2\alpha}$$

$$dN = n d\Omega \frac{dE}{2\alpha}$$

$$g = \frac{1}{2} \left(\frac{\hbar^2}{2m} \left(\frac{2\pi}{L} \right)^2 \right)^{-3/2} \sqrt{E} d\Omega$$

$$g = \frac{dN}{dE}$$

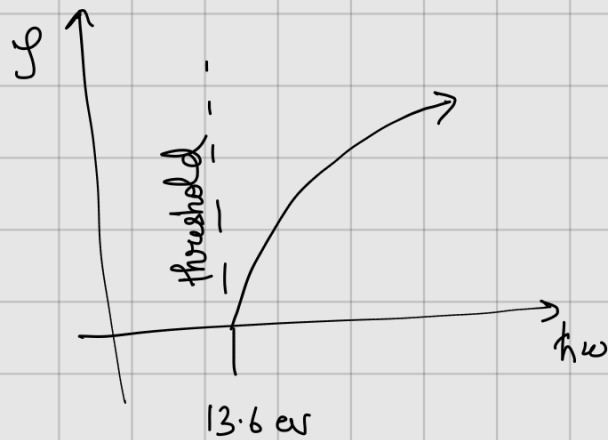
$$g = \frac{L^3}{2} \left(\frac{\hbar^2}{2m} \right)^{-3/2} (2\pi)^{-3} \sqrt{E} d\Omega$$

$g \propto L^3 \leftarrow$

$\therefore$ if we make our box infinite, then the density of states is infinite.

$$W_{i \rightarrow f} = \frac{2\pi}{\hbar} |V_F^+|^2 g(E_i + \hbar\omega) \propto \sqrt{E_i + \hbar\omega}$$

$$GND: \sqrt{\hbar\omega - 13.6 \text{ eV}}$$



V^+ → electric dipole approximation

$$\vec{E} = E_0 \hat{\epsilon} e^{-i\omega t} + E_0^* \hat{\epsilon} e^{i\omega t}$$

$$V = -\vec{d} \cdot \vec{E}(t) = V^+ e^{-i\omega t} + V^- e^{i\omega t}$$

$$V^+ = e E_0 (\vec{r} \cdot \hat{\epsilon})$$

Matrix Element

$$V_{fi}^+ = \langle \vec{k}_f | V^+ | i \rangle = e E_0 \langle \vec{k}_f | \vec{r} \cdot \hat{\epsilon} | i \rangle$$

$$= e E_0 \int_{\substack{d^3 \vec{r} \\ L^3 \rightarrow \infty}} \frac{1}{\sqrt{L^3}} e^{-i \vec{k}_f \cdot \vec{r}} (\vec{r} \cdot \hat{\epsilon}) \psi_i(\vec{r})$$

L^3 can be treated as integral over all space.

$$\propto \frac{1}{\sqrt{L^3}}$$

$$|V_{f_1}^+|^2 \propto \frac{1}{L^3} \rightarrow \text{cancels out } L^3 \text{ in } \int$$

to actually evaluate the value of the integral:

Feynmann's trick

$$\frac{\partial}{\partial k^0} \int d^3\vec{r} e^{-i\vec{k} \cdot \vec{r}} = \int d^3\vec{r} (-i r^0) e^{-i\vec{k} \cdot \vec{r}}$$

$$i \vec{\nabla}_{\vec{k}} \int d^3\vec{r} e^{-i\vec{k} \cdot \vec{r}} = \int d^3\vec{r} e^{-i\vec{k} \cdot \vec{r}} \vec{r}$$

